DRAGER EVITA V800 NEONATAL VENTILATOR SET UP (VN800)

S.O.P



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	Version 1.0	Implementation Guide

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1. Introduction

Dräger has developed software that will allow Evita V800 ventilators to provide mechanical ventilation to patients under 5 kilograms in weight. This software has been approved and downloaded onto ventilators within PICU. This device provides modes for mandatory ventilation and for supporting spontaneous breathing.

Our unit will have 10 ventilators with this specific software update installed. Although this software has been downloaded onto existing equipment, there are a few discrepancies with the set up and use of the ventilator circuit whilst operating within the neonatal mode.

2. Purpose of this Guide

This guide will provide you with information to correctly assemble, operate, dismantle and troubleshoot the Dräger Evita V800 with Neonatal Software.

There will be a step by step guide for:

- Selecting the correct ventilator circuit, humidification delivery and patient category.
- Carrying out the appropriate safety checks.
- Troubleshooting common error messages.
- Replacement of equipment.
- Dismantling the ventilator appropriately, correctly packaging equipment requiring sterilisation and documentation.

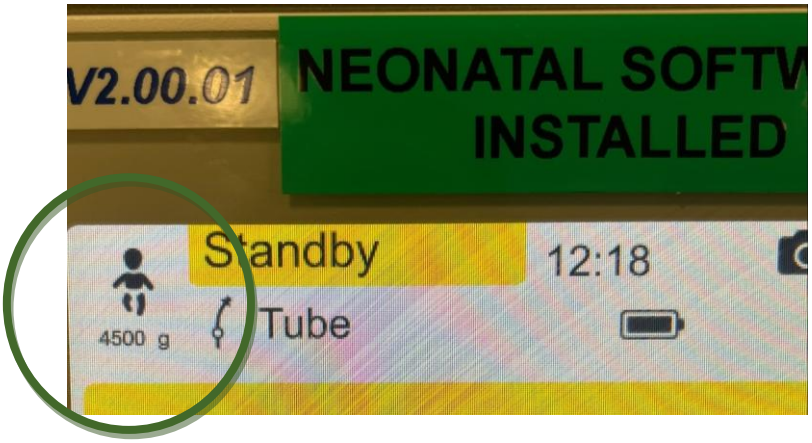
3. Selecting Correct Patient Demographic and Equipment

The **Green Pathway** distinguishes between Evita V800 and Evita V800 ventilators with neonatal capabilities (VN800). When mechanically ventilating a patient less than 5kgs in PICU you **must** use the Evita V800 with neonatal software.

The **Green Pathway** is also to signify the reusable parts of the neonatal kit.

Several components of the ventilator are able to be dismantled and disinfected via the CSSD.

The picture below identifies when the ventilator is in neonatal mode and **Green Pathway** labelling:



Ventilator circuit selection:

Intersurgical® humidified paediatric 15mm dual limb ventilator circuit for VN800. Current paediatric circuit used for patients over 5kgs.

4. Stages of Assembly

1. Select a Dräger Ventilator with neonatal capabilities (Green Labelling) as pictured above.

2. Collect pack containing sterilised “PICU Expiratory Block 5 Parts”, neonatal flow sensor insert and Intersurgical® paediatric ventilator circuit (15mm).





3. Assemble expiratory block within sterile blue tray. Assemble neonatal flow sensor with insert.

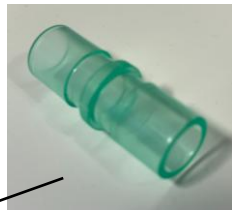


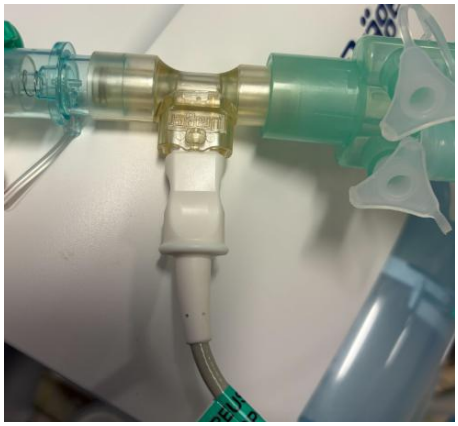




4. Assemble ventilator with circuit and reassembled neonatal expiratory port and flow sensor. **Do not** throw away excess connectors in ventilator circuit pack as connectors will be needed for set-up alongside Intersurgical® connector from expiratory block to expiratory limb. Documentation attached to ventilator must be filled out at this stage.









5. Attach the Dräger neonatal lung to the circuit with an **antibacterial filter** as this will be needed for the pre-use safety check whilst maintaining sterility. Do not use the neonatal lung without a filter as they are not approved for the sterilisation process at present.

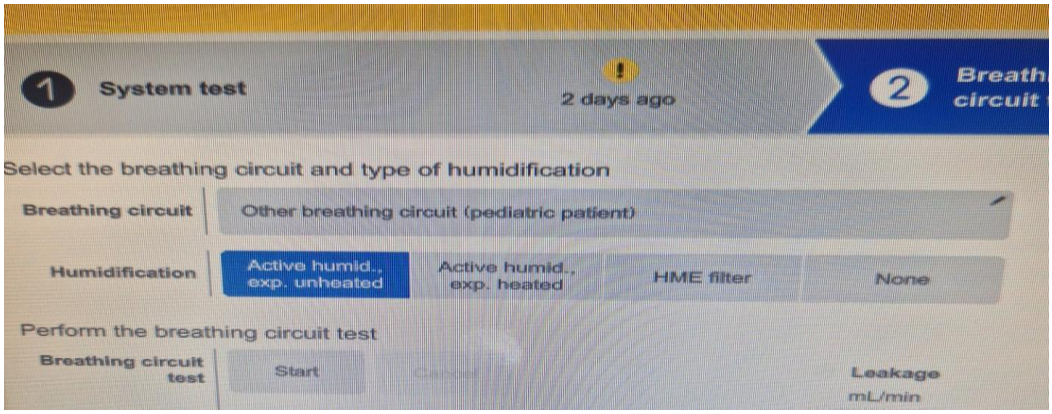
Do not dispose of these lungs after use.

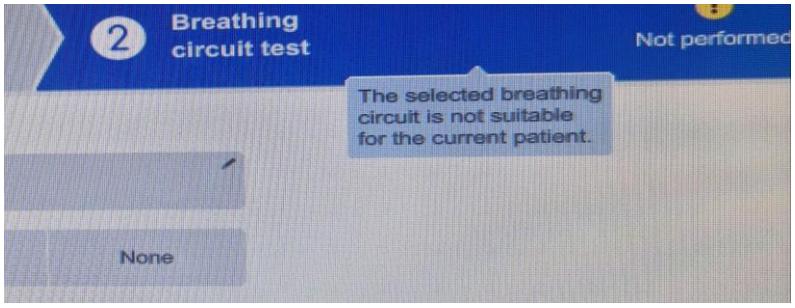




6. The correct circuit and humidification should be selected prior to running through the pre-use safety checks. There can be several warning bubbles that are shown during the set up process.

> Circuit Selection – Within the breathing circuit tab there is a drop down menu used for circuit selection. The circuit that should be selected is **"Other breathing circuit (paediatric patient)"**. Warning bubbles may be displayed. This is due to the ventilator mode and circuit being different (neonatal and paediatric). This is a default safety feature and the paediatric circuit has been deemed safe for use.





> Humidification Selection – The setting for humidification should be **"Active humidification exp. Unheated**". This is set at the default setting and should not require to be changed but must always be checked prior to use

The head of the bed must be positioned to **30 degrees** to ensure the ventilator circuit is dependent from the ETT. Rain out must drain away from the patient.

5. Neonatal Flow Calibration.

This is required to be done prior to first use and every 24 hours when in use. The flow sensor cannot be in use during the calibration process. The calibration process could take place when the patient if off of the ventilator for procedures such as suctioning.

Neonatal flow monitoring

Calibrating the neonatal flow sensor

Performing the calibration, open the dialog page

1. Touch the **Sensors** button in the main menu bar.
2. Touch the **Neonatal flow sensor tab**.



3. Select the sensor type used in the line **Sensor type**.
4. Touch the **Start** button.

Perform the instructions displayed in the tooltip.

5. Confirm the instructions with the **rotary knob**.

The following neonatal flow sensors are available:



Flow sensor ISO 15

Perform the instructions displayed in the tooltip:

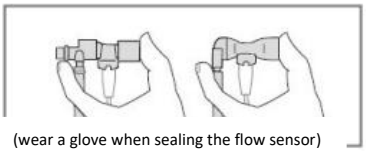
- Remove the tube connector
- Seal the flow sensor

5. Start the calibration by pressing the **rotary knob**.

The tooltip displays information.

After the calibration is complete, the **Start** button turns blue.

After calibration of the neonatal flow sensor connect the tube connector.



(wear a glove when sealing the flow sensor)

Sealing the neonatal flow sensor (wear a sterile glove when sealing the flow sensor)

6. Cleaning Process

The stripping and cleaning process for VN800 differs from the V800. All staff are responsible for checking the correct assembly and dismantling of equipment is carried out.

The expiratory block alongside the neonatal flow sensor can be reused and processed through the sterilisation process.





The equipment that requires to be sterilised can be separated into **5 parts.** Please see images below. Images also include how these parts should be packed prior to leaving PICU for cleaning.

Traceability for these parts is very important.



5 PARTS FROM STRIPPED VENTILATOR



TRAY READY FOR CSSD **GREEN PATHWAY**

Stripping ventilator after use.

The expiratory block does not need to be changed with weekly circuit changes however it must be changed in between patients.

1. Remove the ventilator circuit and dispose.

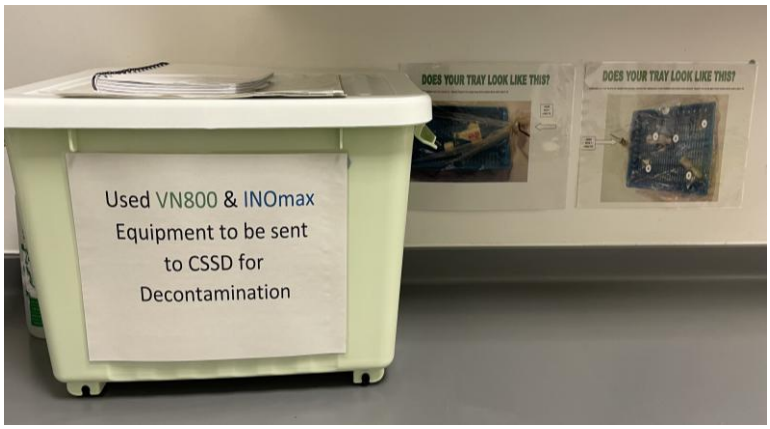
2. The expiratory block **AND** neonatal flow sensor must be removed, dismantled and sent for cleaning to CSSD. Flow sensor insert can be disposed of but flow sensor housing **MUST** be kept.

3. Every neonatal ventilator will have a blue tray with orange barcode attached. **All 5 parts** of the neonatal setup must be sent to CSSD as per photographs below (swan neck with cable tie required). A green bin is located in the sluice for the wrapped trays.











9. CSSD pathway. Once trays are packaged appropriately, PICU healthcare support workers will then deliver trays for cleaning to theatres and collect clean trays for PICU use. Documentation is attached to all neonatal ventilators and must be completed by staff assembling and dismantling equipment.

7. Troubleshooting

Flow Sensor

- Issues with minute, inspiratory and expiratory volumes may occur from inaccurate readings originating from a contaminated or uncalibrated flow sensor.
- The neonatal flow sensor is affected by condensation and water droplets. These sensors are hot-wire anemometers and if saturated by moisture will deliver inaccurate measurements. Removing water or replacing insert to flow sensor and recalibrating should resolve inaccuracy issues.
- Try each or all of these steps if flow sensor appears inaccurate. **Firstly**, calibrate sensor. **Secondly**, change insert of flow sensor and calibrate. **Thirdly**, replace entire flow sensor and calibrate.
- Medical physics have noted potential calibration/safety check failures relating to excessive movement between the flow sensor insert and cable when attached. During testing this was noted to result in failure, apnoea and flow disturbances alarms. If staff note frequent alarms and other troubleshooting methods are not resolving the issue, the cable must be exchange and the original given to the medical physics team.



Apparent rocking back and forth between these parts when the flow sensor is fully assembled and in use with alarms present qualifies as a fault.

Retain previous flow sensor and keep with ventilator so all parts requiring cleaning are sent as a complete kit.

Nitic Oxide

- It has been noted when administering Nitric Oxide therapy in conjunction with the VN800, readings of NO being delivered to the patient can vary outside the accepted 10% range.

  This may be in relation to the higher levels of condensation present disturbing the sampling line or the proximity of the sample line in the circuit to the patient. Current practice of replacing sampling line from iNOMax device to ventilator circuit at an appropriate time improves this issue.

  Replacing the flow sensor insert may also be a solution to this issue.

Condensation

- High volumes of “rain out” have been noted in the circuits when the neonatal mode is in use. Increasing the frequency of yellow bacterial filter changes will help with this issue.

Nebulisers

- Do not run nebulisers through the VN800 set up. The flow sensor membrane is easily blocked and this will prevent the flow sensor from working correctly. The flow sensor will need to be replaced. Nebulisers should be bagged in by hand.

8. Definitions and Abbreviations

VN800 – Dräger Evita V800 Ventilator with Neonatal Software capabilities.

CSSD – Central Sterile Services Department

NO – Nitric Oxide

15mm – 15 millimetre in reference to width of ventilator circuit

9. References

- Drägerwerk AG & Co. KGaA – Information sourced from manufactures recommendations and training.
- Davidson, LM and Berkelhamer, SK (2017). 'Bronchopulmonary Dysplasia: Chronic Lung Disease of Infancy and Long-Term Pulmonary Outcomes', *Journal of Clinical Medicine*. 6(1):4. Available at: doi: 10.3390/jcm6010004.

Please speak to PICU Education Team for further resources.